

AP-E8 Topology-Preserving Lattice Action: An Unanchored Finite-Grid $B = 1$ Stationary Family with a Fail-Closed Continuum Boundary

Route F research note

17 July 2026

Abstract

AP-E7 proved that the unrestricted fixed-boundary finite-site space is connected, so the naive sampled action has no exact integer topological sectors and its unanchored relaxations unwind. This note changes that specific blocker. On a fixed Freudenthal triangulation we add a logarithmic barrier to every unique pair of vertices sharing a tetrahedron. Its linear term is subtracted, so it diverges at a pairwise admissibility floor that prevents reaching the dislocation locus while contributing only $O(a^2)$ to a uniformly smooth sampled field. Pairwise admissibility gives an explicit 0.50744 lower bound on the norm of every affine tetrahedron. The normalized interpolation is therefore nonsingular, its degree is locally constant, and no continuous finite-energy path changes B .

For every fixed grid and every nonempty admissible component, compactness of the sphere variables, coercivity of the breathing field, and divergence of the barrier prove existence of an interior unanchored minimizer. Sampling a smooth continuum degree-one field shows that such $B = 1$ components exist on all sufficiently fine grids. Numerically, seven independently re-relaxed representatives at four spacings and three volumes have direct full tangent gradient densities below 8×10^{-7} , three-target charge $(1, 1, 1)$, and positive barrier margins without a core or centre pin. Nearby barrier parameters give the same finite-grid conclusion.

The result is deliberately narrower than a continuum theorem. The grid-indexed minimizers have not been proved equicoercive, convergent, barrier-independent, or triangulation-independent. No physical two-colour gauge measure, determinant Hessian, BRST superdeterminant, HMC calculation, or degree-one Route-E portal is supplied. The finite-grid topology/stationarity sub-blocker is closed; the physical Lane-B conjunction remains open.

Contents

1	Decision and logical scope	2
2	Finite-grid data and the AP-E7 base action	2
3	The topology-preserving action	3
4	No-zero lemma and exact finite-energy sectors	3
5	Existence of an unanchored finite-grid stationary family	4
6	Smooth-field scaling of the new term	5
7	Exact first and second variations	5
8	Numerical protocol	6

9 Numerical results	7
9.1 Four spacings and three volumes	7
9.2 Barrier-parameter controls	7
9.3 Compact-supported smooth-profile scaling	7
10 Machine gates and changed blocker	8
11 Ordered next work	8
12 Reproduction	9

1 Decision and logical scope

The three proposed blocker-changing paths were:

- (i) complete a finite Ginsparg–Wilson/domain-wall target kernel;
- (ii) replace the naive lattice action by a topology-preserving one and find an unanchored stationary sequence;
- (iii) compute the actual $SO(3)$ Yukawa mapping-torus mod-two index.

The first route still lacks one target mass-family lift, its mixed determinant orientation, and an all-scale $k = +2$ mechanism. The third route can select a Pfaffian torsion line but does not repair the finite-site unwinding found in AP-E7. Route (ii) is selected because it directly changes the proved finite-site obstruction: the infinite barrier removes the dislocation locus from the finite-energy configuration space and creates separated components.

This construction is inspired by geometric lattice charge and lattice Skyrmion admissibility ideas [1, 2, 3, 4], but the specific action below uses a subtracted logarithm and counts every unordered Freudenthal co-cell pair only once. Its finite-energy boundary lies strictly inside the normalized-affine admissible region, before any true dislocation. No claim of regulator uniqueness is made. Recent dense two-colour effective theories show that diquark, scalar, vector, and axial channels can all affect the dynamics [5, 6]; they do not convert the present three-dimensional classical regulator into four-dimensional QC2D.

2 Finite-grid data and the AP-E7 base action

Let $\Lambda_{a,L}$ be an N^3 cubic grid with spacing $a = L/(N - 1)$. The outer faces are fixed to

$$n_x = e_0 = (1, 0, 0, 0), \quad s_x = 1, \quad x \in \partial\Lambda_{a,L}. \quad (2.1)$$

At each of the $M = (N - 2)^3$ interior sites, $n_x \in S^3 \subset \mathbb{R}^4$ and $s_x \in \mathbb{R}$. Before admissibility, the configuration space is

$$\mathcal{C}_N = (S^3)^M \times \mathbb{R}^M, \quad (2.2)$$

which is path-connected.

The AP-E7 base action is retained literally:

$$\begin{aligned}
E_a[n, s] = & \frac{a}{2} \sum_{\langle xy \rangle} (\|n_y - n_x\|^2 + (s_y - s_x)^2) \\
& + \frac{\kappa a^3}{2} \sum_c \sum_{i < j} (\|u_i^c\|^2 \|u_j^c\|^2 - (u_i^c \cdot u_j^c)^2) \\
& + a^3 \sum_{x \in \text{int}} \left[\frac{m_s^2}{2} (s_x - 1)^2 + \beta^2 s_x^2 (1 - n_{x0}) \right], \quad (2.3)
\end{aligned}$$

where

$$u_i^c = \frac{n_{c+i} - n_c}{a}, \quad \kappa = 1, \quad \beta = \frac{1}{2}, \quad m_s = \frac{5}{2}. \quad (2.4)$$

Every term in Eq. (2.3) is nonnegative. The last line is coercive in all interior s_x . This is a classical Skyrme-type regulator [7], not a lattice gauge-theory action.

Each cube is split into the six Freudenthal tetrahedra. For a permutation $p \in S_3$, their vertices are

$$x_0 = c, \quad x_1 = c + e_{p_1}, \quad x_2 = c + e_{p_1} + e_{p_2}, \quad x_3 = c + (1, 1, 1). \quad (2.5)$$

Let $\mathcal{E}_F$ be the set of *unique unordered* vertex pairs $\{x, y\}$ that co-occur in at least one such tetrahedron. Pair incidence is not multiplied when the same edge belongs to several tetrahedra.

3 The topology-preserving action

Fix

$$-\frac{1}{3} < \epsilon < 1, \quad \gamma > 0, \quad q_{xy} = n_x \cdot n_y. \quad (3.1)$$

For the production card, $\epsilon = 0.01$ and $\gamma = 1$. Put

$$b_\epsilon(t) = \begin{cases} -\log \frac{t-\epsilon}{1-\epsilon} + \frac{t-1}{1-\epsilon}, & t > \epsilon, \\ +\infty, & t \leq \epsilon. \end{cases} \quad (3.2)$$

The AP-E8 action is

$$\boxed{E_{a,\epsilon,\gamma}^{\text{TP}}[n, s] = E_a[n, s] + \gamma a \sum_{\{x,y\} \in \mathcal{E}_F} b_\epsilon(q_{xy}).} \quad (3.3)$$

The subtraction in Eq. (3.2) is essential. With $d = 1 - \epsilon$, direct differentiation gives

$$b_\epsilon(1) = b'_\epsilon(1) = 0, \quad b'_\epsilon(t) = -\frac{1}{t-\epsilon} + \frac{1}{d} = \frac{t-1}{d(t-\epsilon)}, \quad (3.4)$$

$$b''_\epsilon(t) = \frac{1}{(t-\epsilon)^2} > 0, \quad \lim_{t \downarrow \epsilon} b_\epsilon(t) = +\infty. \quad (3.5)$$

Since $t \leq 1$ for unit vectors, b_ϵ decreases from infinity to zero and is nonnegative on its finite domain. Writing $u = 1 - t \in [0, d)$,

$$b_\epsilon(1-u) = -\log \left(1 - \frac{u}{d}\right) - \frac{u}{d} = \sum_{k=2}^{\infty} \frac{1}{k} \left(\frac{u}{d}\right)^k = \frac{u^2}{2d^2} + O(u^3). \quad (3.6)$$

Thus no term linear in $1 - t$ remains.

4 No-zero lemma and exact finite-energy sectors

Lemma 4.1 (Pairwise admissibility implies a uniform no-zero bound). *Let $n_0, \dots, n_3 \in S^3$ obey $n_i \cdot n_j > \epsilon$ for every $i \neq j$. For all barycentric coordinates $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$,*

$$\left\| \sum_{i=0}^3 \lambda_i n_i \right\|^2 \geq \epsilon + (1-\epsilon) \sum_{i=0}^3 \lambda_i^2 \geq \frac{1+3\epsilon}{4} > 0. \quad (4.1)$$

Proof. Using $2 \sum_{i < j} \lambda_i \lambda_j = 1 - \sum_i \lambda_i^2$,

$$\begin{aligned} \left\| \sum_i \lambda_i n_i \right\|^2 &= \sum_i \lambda_i^2 + 2 \sum_{i < j} \lambda_i \lambda_j (n_i \cdot n_j) \\ &\geq \sum_i \lambda_i^2 + 2\epsilon \sum_{i < j} \lambda_i \lambda_j = \epsilon + (1-\epsilon) \sum_i \lambda_i^2. \end{aligned} \quad (4.2)$$

The first inequality is strict whenever at least two barycentric weights are positive; at a vertex both sides of that comparison equal one. Finally, the Cauchy–Schwarz inequality gives $\sum_i \lambda_i^2 \geq 1/4$. The last bound is positive exactly when $\epsilon > -1/3$. $\square$

For $\epsilon = 0.01$, Eq. (4.1) yields

$$\min_T \min_{\lambda \in \Delta^3} \left\| \sum_i \lambda_i n_i \right\| \geq \sqrt{\frac{1+3\epsilon}{4}} = 0.507444578255 \dots \quad (4.3)$$

Therefore the normalized affine interpolation

$$\hat{n}_T(\lambda) = \frac{\sum_i \lambda_i n_i}{\left\| \sum_i \lambda_i n_i \right\|} \quad (4.4)$$

is continuous on every tetrahedron, including common faces.

Define the admissible open set

$$\mathcal{A}_{\epsilon,N} = \{(n, s) \in \mathcal{C}_N : q_{xy} > \epsilon \text{ for all } \{x, y\} \in \mathcal{E}_F\}. \quad (4.5)$$

Collapsing the vacuum boundary to a point turns Eq. (4.4) into a continuous map $S^3 \rightarrow S^3$. Its degree $B_{\text{geom}} \in \mathbb{Z}$ is constant on every connected component of $\mathcal{A}_{\epsilon,N}$.

Theorem 4.2 (Finite-energy sector preservation). *Every continuous path with finite $E_{a,\epsilon,\gamma}^{\text{TP}}$ remains in one component of $\mathcal{A}_{\epsilon,N}$, and its normalized-affine degree is constant. Consequently no continuous finite-energy path joins $B = 1$ to the vacuum.*

Proof. Finite action requires every $q_{xy} > \epsilon$, so the entire path lies in $\mathcal{A}_{\epsilon,N}$. A connected path cannot change connected component. Equation (4.1) makes the normalized interpolation a continuous homotopy along the path, and homotopy invariance fixes its degree. Equivalently, any route to another degree must meet some $q_{xy} = \epsilon$, where Eq. (3.2) has infinite energy. $\square$

5 Existence of an unanchored finite-grid stationary family

Theorem 5.1 (Fixed-grid minimizer). *For fixed $a > 0$, fixed finite N , $\gamma > 0$, $m_s > 0$, and any nonempty connected component $K \subset \mathcal{A}_{\epsilon,N}$, the action $E_{a,\epsilon,\gamma}^{\text{TP}}$ attains a minimum at an interior point of K . That minimizer is stationary under every interior S^3 -tangent variation of n and every interior variation of s . No core or centre anchor is required.*

Proof. Choose a minimizing sequence in K . After discarding finitely many terms, its energy is bounded by a finite constant C . Nonnegativity of every term in Eq. (3.3) implies, for every edge,

$$b_\epsilon(q_{xy}) \leq \frac{C}{\gamma a}. \quad (5.1)$$

The strictly decreasing map $b_\epsilon : (\epsilon, 1] \rightarrow [0, \infty)$ is invertible. Hence there is a fixed-grid $\delta_{a,C} > 0$ such that

$$q_{xy} \geq \epsilon + \delta_{a,C} \quad (5.2)$$

for every member of the sequence and every edge. All n_x already lie in the compact finite product $(S^3)^M$. The mass term gives

$$\frac{a^3 m_s^2}{2} \sum_{x \in \text{int}} (s_x - 1)^2 \leq C, \quad (5.3)$$

so the finite vector (s_x) is bounded. A subsequence converges. The uniform margin (5.2) keeps its limit inside $\mathcal{A}_{\epsilon,N}$; connected components are closed relative to that open set, so the limit remains in K . Continuity on the finite-energy domain shows that the limit attains the infimum.

The margin also supplies a neighborhood contained in the same admissible component. The minimizer is therefore an unconstrained interior critical point of the fixed-boundary finite-dimensional manifold. Its tangent and scalar first variations vanish. No value at an interior site was fixed in the argument. $\square$

Theorem 5.2 (Nonempty $B = 1$ components on a refining grid sequence). *Let $n : \bar{\Omega} \rightarrow S^3$ be a smooth degree-one map equal to the vacuum near $\partial\Omega$, and let s be smooth. For any fixed $\epsilon < 1$, all sufficiently fine Freudenthal grids obtained by sampling (n, s) belong to $\mathcal{A}_{\epsilon, N}$ and their normalized-affine degree is one. Consequently Theorem 5.1 supplies an unanchored stationary minimizer (n_a, s_a) in a $B = 1$ component for every grid in a sequence $a \downarrow 0$.*

Proof. Uniform continuity of n on the compact domain implies that the images of any two vertices sharing a tetrahedron approach one another uniformly as $a \rightarrow 0$. Their dot products therefore approach one uniformly and exceed ϵ for sufficiently small a . The piecewise affine interpolation of the sampled values converges uniformly to n , and its norm converges uniformly to one. After normalization it is uniformly close to n ; for sufficiently small a , the pointwise shortest-geodesic homotopy is well-defined. Thus its degree equals $\deg n = 1$. Apply the fixed-grid theorem to the component containing that sample. $\square$

Remark 5.3 (What “sequence” does and does not mean). The theorem gives a family of exact finite-dimensional minimizers indexed by the grid spacing. It does not give a compactness estimate uniform in a . Indeed, Eq. (5.1) permits $\delta_{a,C}$ to shrink rapidly as $a \downarrow 0$. The theorem alone does not imply that (n_a, s_a) converges, that its limit solves a continuum Euler–Lagrange equation, or that a barrier-supported lattice core cannot form.

6 Smooth-field scaling of the new term

Let an edge have physical displacement av with v fixed by the Freudenthal stencil. For a uniformly smooth unit field,

$$n(x) \cdot n(x + av) = 1 - \frac{a^2}{2} \|D_v n(x)\|^2 + O(a^3). \quad (6.1)$$

Combining Eqs. (3.6) and (6.1),

$$b_\epsilon(n(x) \cdot n(x + av)) = \frac{a^4}{8(1 - \epsilon)^2} \|D_v n(x)\|^4 + O(a^5). \quad (6.2)$$

There are $O(a^{-3})$ unique edges at fixed physical volume. Therefore

$$\gamma a \sum_{\{x,y\} \in \mathcal{E}_F} b_\epsilon(q_{xy}) = \frac{\gamma a^2}{8(1 - \epsilon)^2} \int_{\Omega} \sum_v \rho_v \|D_v n\|^4 d^3x + O(a^3), \quad (6.3)$$

where ρ_v are the direction/incidence densities of the chosen unique-edge stencil. Thus the new term has no leading kinetic renormalization and is $O(a^2)$ on a fixed uniformly smooth family.

Equation (6.3) is not uniform over the unknown minimizers. It proves neither equicoercivity nor Gamma convergence. It also does not remove finite- a triangulation dependence, because the ρ_v depend on the chosen stencil.

7 Exact first and second variations

Let $w = \gamma a$, $q = n_x \cdot n_y$, and consider one barrier edge $w b_\epsilon(q)$. Its ambient gradients are

$$G_x^{(xy)} = w b'_\epsilon(q) n_y, \quad G_y^{(xy)} = w b'_\epsilon(q) n_x. \quad (7.1)$$

The ambient Hessian blocks are

$$\begin{aligned} H_{xx}^{(xy)} &= w b''_\epsilon(q) n_y n_y^T, & H_{yy}^{(xy)} &= w b''_\epsilon(q) n_x n_x^T, \\ H_{xy}^{(xy)} &= w [b''_\epsilon(q) n_y n_x^T + b'_\epsilon(q) I_4], & H_{yx}^{(xy)} &= (H_{xy}^{(xy)})^T. \end{aligned} \quad (7.2)$$

For tangent vectors $\xi_x \perp n_x$, $\xi_y \perp n_y$, take the product geodesic on $S^3 \times S^3$. Its first dot-product variation is

$$\delta q = \xi_x \cdot n_y + n_x \cdot \xi_y, \quad (7.3)$$

and its second variation is

$$\delta^2 q = 2\xi_x \cdot \xi_y - q(\|\xi_x\|^2 + \|\xi_y\|^2). \quad (7.4)$$

Hence the exact intrinsic edge quadratic form is

$$\begin{aligned} \delta^2(wb_\epsilon(q)) &= w \left\{ b''_\epsilon(q)(\xi_x \cdot n_y + n_x \cdot \xi_y)^2 \right. \\ &\quad \left. + b'_\epsilon(q) [2\xi_x \cdot \xi_y - q(\|\xi_x\|^2 + \|\xi_y\|^2)] \right\}. \end{aligned} \quad (7.5)$$

For the full action, let $G_x = \partial E^{\text{TP}} / \partial n_x$, $P_x = I_4 - n_x n_x^T$, and $\lambda_x = n_x \cdot G_x$. Stationarity is

$$P_x G_x = 0, \quad \frac{\partial E^{\text{TP}}}{\partial s_x} = 0. \quad (7.6)$$

If Q collects orthonormal tangent frames and the scalar directions, the full Riemann Hessian at a stationary point is

$$H_{\text{Riem}} = Q^T [H_{\text{emb}} - \text{diag}(\lambda_1 I_4, 0, \lambda_2 I_4, 0, \dots)] Q. \quad (7.7)$$

Projecting H_{emb} without the sphere-curvature shift is incorrect. The AP-E8 card validates Eq. (7.5) independently; it does not assemble or promote Eq. (7.7) as a physical gauge–meson–ghost–fermion super-Hessian.

8 Numerical protocol

The reproducible implementation is

`route_f/code/scan_ap_e8_topology_preserving_b1.py`.

It imports the canonical AP-E6 continuum profile only as initial data. The little-endian (r, F, s) checksum is

81104f59a5f3f4337739fc2a217cafca8da91581c9f1fb40b4fdb6ce0f948d59.

Every grid receives exact vacuum boundary data and is minimized independently. No core, centre, or interior field value is fixed.

A fixed-base exponential chart is used within each L-BFGS solve and is re-centred if needed. An invalid trial point receives a solver-only continuation which matches the exact b, b', b'' at a positive cutoff and adds a differentiable quartic rejection below the admissibility floor. Its energy/gradient pair is checked separately. All reported fields are re-evaluated against the literal infinite-barrier domain. Because the complete geodesic segment of each internal quasi-Newton line search was not interval-certified, `complete_line_search_segments_certified_admissible=false`. This does not affect the final-point stationarity test or the existence theorem, but it forbids a stronger claim about the numerical trajectory.

At every final field the code discards the chart derivative and recomputes the direct projected gradient

$$g_a = \frac{\left[\sum_x \|P_x G_x\|^2 + \sum_x (\partial E^{\text{TP}} / \partial s_x)^2 \right]^{1/2}}{a^3 \sqrt{4M}}. \quad (8.1)$$

The stationarity threshold is $g_a < 2 \times 10^{-6}$. The degree is counted independently with three generic regular targets. A size diagnostic is

$$R_{\text{rms}}^2 = \frac{\sum_x (1 - n_{x0}) |x|^2}{\sum_x (1 - n_{x0})}. \quad (8.2)$$

It is a profile diagnostic, not a proof of localization or continuum compactness.

Three independent calculus regressions precede the relaxation:

$$\begin{aligned} \text{total-gradient directional residual} &= 9.38 \times 10^{-11}, \\ \text{invalid-continuation gradient residual} &= 2.81 \times 10^{-11}, \\ \text{barrier geodesic-Hessian residual} &= 8.16 \times 10^{-9}. \end{aligned} \tag{8.3}$$

9 Numerical results

9.1 Four spacings and three volumes

The seven principal solutions are:

N	L	a	E^{TP}	g_a	$\min q$	B rows	R_{rms}/a
15	6	0.428571	75.090097547	1.651×10^{-7}	0.091845	(1, 1, 1)	3.0704
17	6	0.375000	72.444298275	3.421×10^{-7}	0.090442	(1, 1, 1)	3.3504
19	6	0.333333	70.831336466	2.028×10^{-7}	0.083178	(1, 1, 1)	3.6277
21	6	0.300000	69.844122881	7.704×10^{-7}	0.077786	(1, 1, 1)	3.8980
16	6	0.400000	73.627393329	2.150×10^{-7}	0.090722	(1, 1, 1)	3.2634
21	8	0.400000	73.346860060	1.052×10^{-7}	0.095418	(1, 1, 1)	3.5780
26	10	0.400000	73.289478682	1.228×10^{-7}	0.095975	(1, 1, 1)	3.8613

All pair margins $\min q - \epsilon$ are positive. The fixed- $a = 0.4$ energy spread over $L = 6, 8, 10$ is

$$\frac{E_{\max} - E_{\min}}{E_{\max}} = 0.00458952. \tag{9.1}$$

The fixed- $L = 6$ physical radii for $N = 15, 17, 19, 21$ are 1.3159, 1.2564, 1.2092, 1.1694, while R_{rms}/a grows. This is evidence against a literal one-cell collapse in the displayed sequence, but it is not a uniform lower-radius theorem.

9.2 Barrier-parameter controls

At $N = 15, L = 6$, four nearby choices give:

γ	ϵ	E^{TP}	g_a	$\min q$	B rows
0.5	0.010	63.665861079	1.614×10^{-7}	0.044767	(1, 1, 1)
2.0	0.010	90.696115305	1.794×10^{-7}	0.221220	(1, 1, 1)
1.0	0.005	74.773915108	1.487×10^{-7}	0.085519	(1, 1, 1)
1.0	0.020	75.719960191	2.790×10^{-7}	0.104399	(1, 1, 1)

These controls show persistence of admissible stationary representatives, not regulator independence: the finite- a energies and profiles visibly depend on γ and ϵ .

A deterministic 2×10^{-3} tangent-plus-scalar perturbation of the $N = 15, L = 6$ initial sample returns to a numerically indistinguishable same-energy admissible stationary representative: its final energy differs from the baseline by less than 10^{-13} relatively, has three-target $B = (1, 1, 1)$, and passes the direct stationarity threshold. This is a local-basin control, not a proof that the reported point is the global minimizer of its component.

9.3 Compact-supported smooth-profile scaling

To avoid a boundary-layer artifact, this regression does not overwrite the nonzero tail of the canonical AP-E6 profile. It uses a separate C^2 , degree-one hedgehog with support radius two inside the $L = 6$ box, so the field already equals the exact vacuum on every boundary face. Its barrier energy is:

N	a	E_{barrier}	$\min q$
17	0.375000	30.439452623	0.150863
21	0.300000	18.075663943	0.360734
25	0.250000	12.263778095	0.551121
29	0.214286	8.914172689	0.656982
33	0.187500	6.784938174	0.722692

A log–log fit gives slope 2.15910 with $R^2 = 0.999141$. The four successive local slopes are 2.3356, 2.1276, 2.0695, 2.0440, approaching the analytic exponent two. The fit is not applied to the re-relaxed minimizers and cannot establish their continuum behavior.

10 Machine gates and changed blocker

The production card passes 13/13 checks. It writes

```
route_f/output/ap_e8_topology_preserving_b1.json,
route_f/output/ap_e8_topology_preserving_b1.md.
```

The following statements are now true:

1. a topology-preserving finite-grid action is explicitly defined;
2. its finite-energy degree is preserved by theorem;
3. every nonempty fixed-grid $B = 1$ component has an unanchored minimizer;
4. such components and minimizers exist on a sufficiently fine grid sequence;
5. seven numerical final fields pass a direct unanchored stationarity test, three-target $B = 1$, and a positive barrier margin.

This changes the AP-E7 conclusion. The former statement “no unanchored admissible finite-grid stationary representative has been found” is superseded. The correct new statement is

$$\boxed{C_{B1}^{\text{finite grid}} = \text{true}, \quad C_{B1}^{\text{physical continuum}} = \text{false}.} \quad (10.1)$$

Fail-closed gate 10.1 (Continuum and physics remain fail closed). None of the following has been established: a numerically certified global minimum, complete line-search-segment admissibility, equicoercivity, Gamma convergence, a continuum stationary limit, barrier or triangulation independence, a physical charged-QC2D action, an interacting gauge–meson–ghost–fermion Hessian, a BRST superdeterminant, four-dimensional dynamics, HMC, or a quantum continuum limit.

Consequently the full Lane-B closure condition remains false and no degree-one Route-E portal is authorized. The two deferred alternatives—a complete finite GW/domain-wall target kernel and the actual $SO(3)$ Yukawa mapping-torus mod-two index—remain independent research lanes.

11 Ordered next work

The next calculations should be ordered by their ability to distinguish a physical continuum object from a barrier artifact:

1. Prove an equicoercive estimate or a Gamma-liminf statement for a controlled scaling $\gamma(a), \epsilon(a)$, or produce a counterexample showing barrier-core collapse. Measure local gradient concentration and a topological radius, not only total energy.

2. Repeat the stationary family for at least two alternative cube triangulations and several $\gamma(a), \epsilon(a)$ trajectories. Extrapolate observables jointly in a, L, γ, ϵ ; do not extrapolate the barrier energy as if it were physical.
3. Only after one regulator-stable continuum background is identified, assemble the full same-action Riemann Hessian including the curvature shift in Eq. (7.7), then add interacting gauge/ghost blocks and the bosonic second variation of one regulated fermion determinant.
4. In parallel, complete either the finite target GW/domain-wall operator or the actual $SO(3)$ mapping-torus mod-two index. A portal remains last.

12 Reproduction

From the repository root, run

```
python3 route_f/code/scan_ap_e8_topology_preserving_b1.py
```

The JSON serializer rejects NaN and infinity. The output records Python, NumPy, SciPy, source SHA-256 values, the canonical seed checksum, every main and control case, the three calculus audits, all Boolean theorem and fail-closed flags, and the 13/13 checks.

References

- [1] B. Berg and M. Luescher. Definition and statistical distributions of a topological number in the lattice $O(3)$ sigma model. *Nuclear Physics B*, 190:412–424, 1981.
- [2] B. Berg. Dislocations and topological background in the lattice $O(3)$ sigma model. *Physics Letters B*, 104:475–480, 1981.
- [3] R. S. Ward. Stable topological skyrmions on the 2-d lattice. *Letters in Mathematical Physics*, 35:385–393, 1995.
- [4] Alec J. Schramm and Benjamin Svetitsky. Topology and metastability in the lattice Skyrme model. *Physical Review D*, 62:114020, 2000.
- [5] Daiki Suenaga, Kotaro Murakami, Etsuko Itou, and Kei Iida. Probing the hadron mass spectrum in dense two-color QCD with the linear sigma model. *Physical Review D*, 107:054001, 2023.
- [6] Daiki Suenaga, Kotaro Murakami, Etsuko Itou, and Kei Iida. Mass spectrum of spin-one hadrons in dense two-color QCD: Novel predictions by extended linear sigma model. *Physical Review D*, 109:074031, 2024.
- [7] T. H. R. Skyrme. A nonlinear field theory. *Proceedings of the Royal Society A*, 260:127–138, 1961.